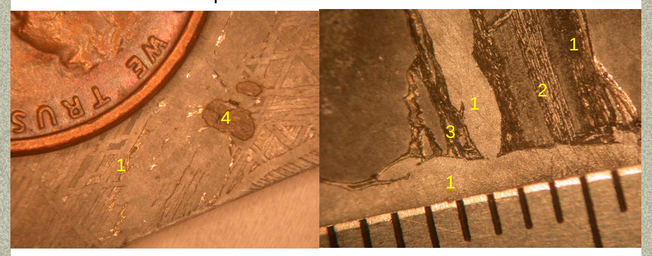
Determining the Mechanical and Thermodynamic Properties of Iron Meteorites Below 300K

Matthew Bonidie[1], Cyril Opeil[1], Robert Macke[2], Guy Consolmagno[3], Daniel Britt[2]
[1]Department of Physics, Boston College
[2]Department of Physics, University of Central Florida
[3]Specola Vaticana, Vatican City State

Iron Meteorites

Iron meteorites are primarily kamacite or taenite based Fe-Ni polycrystalline alloys that solidify from planetary cores over a timespan of $t \geq 10^6$ years via thermal radiation into the vacuum of space. Depending on their Ni composition and cooling rates, iron meteorites crystallize into either fine octahedrite, coarse octahedrite, or hexahedrite crystal structures. Several phases of meteorite are shown below:



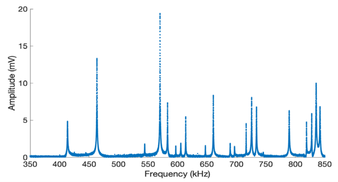
Legend: 1. Taenite, 2. Kamacite, 3. Plessite Pools, 4. Inclusion

RUS Technique

RUS utilizes the probing of the resonant modes of a solid single crystal or polycrystal material in order to determine the complete elastic stiffness matrix for that given material.

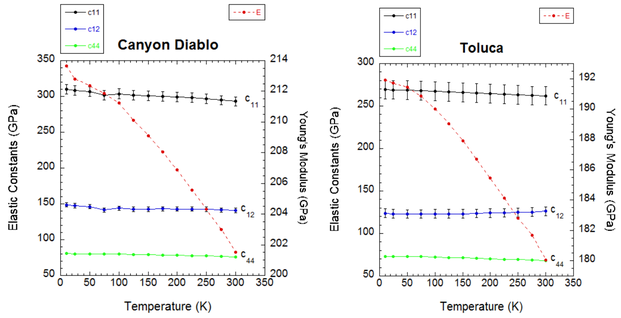
$$\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \gamma_{23} \\ \gamma_{31} \\ \gamma_{12} \end{bmatrix} = \begin{bmatrix} C_{11} & C_{12} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{11} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{12} & C_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & 2(C_{11}-C_{12}) & 0 & 0 \\ 0 & 0 & 0 & 0 & 2(C_{11}-C_{12}) & 0 \\ 0 & 0 & 0 & 0 & 0 & 2(C_{11}-C_{12}) \end{bmatrix} \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \tau_{23} \\ \tau_{31} \\ \tau_{12} \end{bmatrix}$$

Samples are prepared by cutting a cubic sample via a diamond saw and polishing to ensure complete orthogonality of each sample face. Samples are then mounted into the RUS probe and place into a Quantum Design PPMS for measurement at varying temperature. Below is a sample RUS scan, as well as an example of a prepared sample:



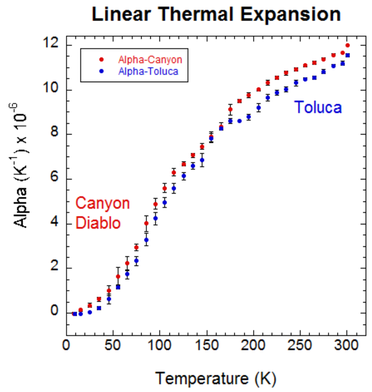
RUS Data

The isotropic elastic constants and Young's Modulus for a coarse (Canyon Diablo) and fine (Toluca) meteorite structure are shown below:



The constants behave approximately linearly through the low temperature range. We see that Young's Modulus is also approximately linear in the higher temperature regime, but exhibits curvature starting at ~50 K.

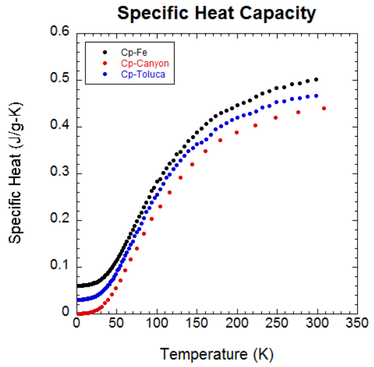
Linear Thermal Expansion



The linear thermal expansion of the two meteorites has also been measured. Due to the inherent disorder of the two meteorites, we can assume that the thermal expansion in any direction of the sample will be the same as the direction we have measured. Thus, a safe estimation of the volumetric thermal expansion can be obtained by simply multiplying the values from the plot above by three.

Specific Heat Capacity

Measurements of each meteorite's specific heat capacity were performed as a fundamental thermodynamic measurement of the material. Note, the two plots have been offset by a factor of 0.03 J/g-K in order to clearly show the curve for both meteorites.



Here, it is important to note the similarities in the two meteorites. Consisting of roughly the same bulk materials, these meteorites are expected to have roughly the same specific heat. Any differences in the material can be primarily attributed to any rocky inclusions in the meteorites.

Conclusions

The crystalline structure of metal meteorites does not seem to impact their overall thermodynamic properties, however, when accounting for their mechanical behavior, coarse or fine grained structures create differences in their elastic stiffness.

Finer structure is due to differences in time of cooling or external parameters while in space and appears to leave the material softer, while leaving its bulk thermodynamic properties unchanged.

Acknowledgements

This work is supported by NASA, SSERVI (Solar System Exploration Research Virtual Institute) and CLASS (Center for Lunar and Asteroid Surface Science) under grant No. NNA14AB05A

Boston College

Society of Physics Students - SPS - Annual Symposium